

Experiment:8

Construct a C program to simulate Round Robin scheduling algorithm with C

Code:-

```
#include <stdio.h>
#include <conio.h>
int main()
{
    int i, NOP, sum = 0, count = 0, y, quant;
    int wt = 0, tat = 0;
    int at[10], bt[10], temp[10];
    float avg_wt, avg_tat;
    printf("Total number of processes: ");
    scanf("%d", &NOP);
    y = NOP;
    for (i = 0; i < NOP; i++)
    {
        printf("\nEnter Arrival Time and Burst Time of Process[%d]\n", i + 1);

        printf("Arrival Time: ");
        scanf("%d", &at[i]);
        printf("Burst Time: ");
        scanf("%d", &bt[i]);
        temp[i] = bt[i];
    }
}
```

```
printf("\nEnter the Time Quantum: ");  
scanf("%d", &quant);  
printf("\nProcess\tBurst Time\tTurnaround Time\tWaiting Time\n");
```

```
for (sum = 0, i = 0; y != 0;)  
{  
    if (temp[i] <= quant && temp[i] > 0)  
    {  
        sum += temp[i];  
        temp[i] = 0;  
        count = 1;  
    }  
    else if (temp[i] > 0)  
    {  
        temp[i] -= quant;  
        sum += quant;  
    }  
}
```

```
if (temp[i] == 0 && count == 1)  
{  
    y--;
```

```
    printf("P%d\t%d\t\t%d\t\t%d\n",  
           i + 1,  
           bt[i],  
           sum - at[i],  
           sum - at[i] - bt[i]);
```

```
        wt += sum - at[i] - bt[i];

        tat += sum - at[i];

        count = 0;
    }
    if (i == NOP - 1)
        i = 0;
    else if (at[i + 1] <= sum)
        i++;
    else
        i = 0;
}

avg_wt = (float)wt / NOP;
avg_tat = (float)tat / NOP;

printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat);

getch();
return 0;
}
```

Output:

```
Total number of processes: 3

Enter Arrival Time and Burst Time of Process[1]
Arrival Time: 0
Burst Time: 5

Enter Arrival Time and Burst Time of Process[2]
Arrival Time: 1
Burst Time: 4

Enter Arrival Time and Burst Time of Process[3]
Arrival Time: 2
Burst Time: 2

Enter the Time Quantum: 2

Process Burst Time      Turnaround Time Waiting Time
P3      2              4              2
P2      4              9              5
P1      5             11              6

Average Waiting Time = 4.33
Average Turnaround Time = 8.00

...Program finished with exit code 0
Press ENTER to exit console.[]
```